ic_autolab round 8 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T04:25:45Z at git `ebe86a2d6dbe856a7e708256cd72450f2b0c5192`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-ff48c0` (`python3 tools/pcc/pcc.py standing PCC-20261104-ff48c0`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
28 **improvement**	ksum4-n18	0.678 [0.666, 0.691]	0.702 [0.687, 0.716]	0.869 [0.841, 0.898]	sel < inc	7.445 / 7.910 / 12.100	10.980 / 11.274 / 13.921	0.956 / 1.045 / 1.848	7.8x / 7.6x / 6.5x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

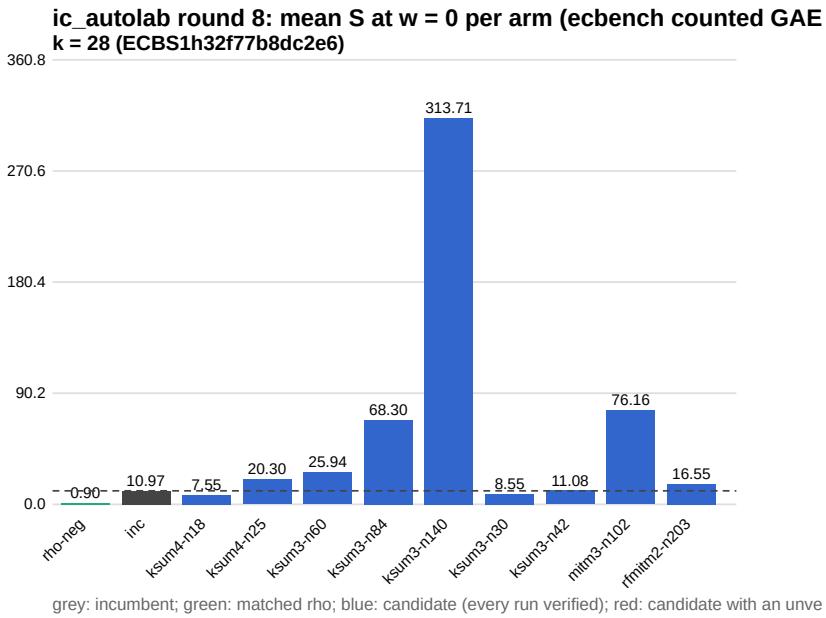
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.6781 [0.666, 0.691] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 28: session `ECBS1h32f77b8dc2e6` (run exit 0, verify exit 0, receipt sha256 `5ea2cf014199d5fda35a3c86bc48f4bd83e1025d43c912b973fe021b5bcaa4b2`)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	32	32	0.897 / 0.981 / 1.730	matched reference
`inc`	ksum4	21	`prime-projected-abscissa:size=21,scan_cap=336`	`ksum:m=4`	32	32	10.972 / 11.266 / 13.912	incumbent
`ksum4-n18`	ksum4	18	`prime-projected-abscissa:size=18,scan_cap=288`	`ksum:m=4`	32	32	7.548 / 8.035 / 12.417	incumbent neighbourhood: N = 0.85 x 21
`ksum4-n25`	ksum4	25	`prime-projected-abscissa:size=25,scan_cap=400`	`ksum:m=4`	32	32	20.301 / 20.483 / 22.122	incumbent neighbourhood: N = 1.2 x 21
`ksum3-n60`	ksum3	60	`prime-projected-abscissa:size=60,scan_cap=960`	`ksum:m=3`	32	32	25.936 / 26.226 / 28.837	cross-size transfer neighbourhood: N = 0.85 x 70 (ksum3 from k = 20)
`ksum3-n84`	ksum3	84	`prime-projected-abscissa:size=84,scan_cap=1344`	`ksum:m=3`	32	32	68.298 / 68.549 / 70.802	cross-size transfer neighbourhood: N = 1.2 x 70

arm	family	N factor base	oracle	runs verified	S (w = 0 / 0.1 / 1)	reason proposed
						(ksum3 from k = 20)
ksum3-n140	ksum3	140	prime-projected-abscissa:size=140,scan_cap=2240	ksum:m=3	32	32	313.708 / 314.199 / 318.622	cross-size transfer neighbourhood: N = 2 x 70 (ksum3 from k = 20)
ksum3-n30	ksum3	30	prime-projected-abscissa:size=30,scan_cap=480	ksum:m=3	32	32	8.550 / 9.586 / 18.909	alternative ksum3 (best S(0) 8.906 at N = 35): N = 0.85 x 35
ksum3-n42	ksum3	42	prime-projected-abscissa:size=42,scan_cap=672	ksum:m=3	32	32	11.082 / 11.591 / 16.166	alternative ksum3 (best S(0) 8.906 at N = 35): N = 1.2 x 35
mitm3-n102	mitm3	102	prime-projected-abscissa:size=102,scan_cap=1632	mitm:m=3,negation_folded=1	32	32	76.155 / 83.702 / 151.618	exploration: family mitm3 untried at k = 28, N = 0.5 x N_a
rfmitm2-n203	rfmitm2	203	prime-projected-abscissa:size=203,scan_cap=3248	rootfind-mitm:m=2	32	32	16.550 / 18.159 / 32.637	exploration: family rfmitm2 untried at k = 28, N = 1 x N_a



train screen

Selection

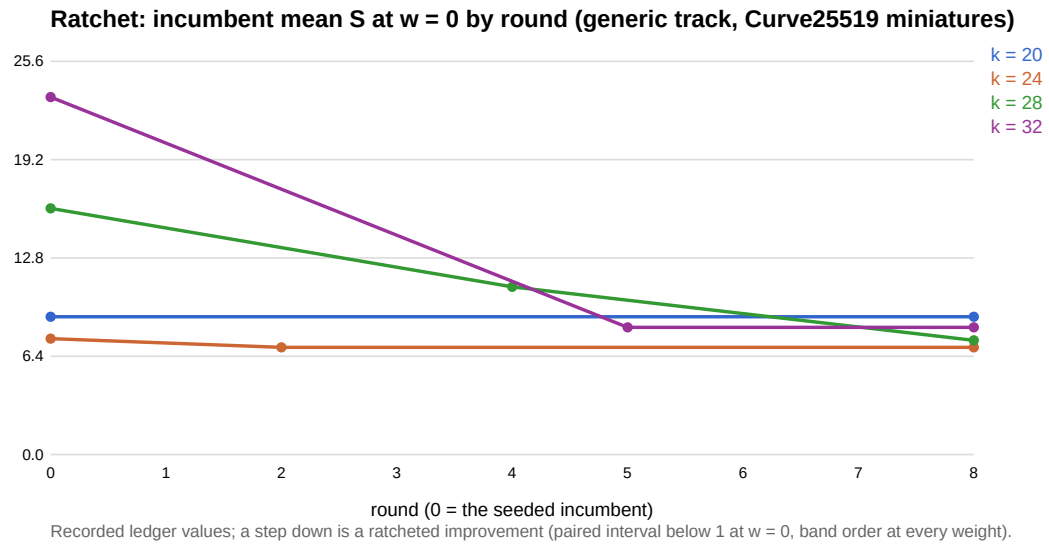
- k = 28: challenger `ksum4-n18` (Pareto alternative at w = 1: `ksum4-n18`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
28	ECBS1hc8e87821dc40	0	0	216709a0d05d32769a88a12df2873aee9ec4b2b94dce51e174bd721710ae2efc	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	ksum3-n11 (prime-projected-abscissa:size=11,scan_cap=176 + ksum:m=3)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	ksum4-n12 (prime-projected-abscissa:size=12,scan_cap=192 + ksum:m=4)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	ksum4-n18 (prime-projected-abscissa:size=18,scan_cap=288 + ksum:m=4)	8	7.445 / 7.910 / 12.100	7.8x / 7.6x / 6.5x
32	ksum4-n26 (prime-projected-abscissa:size=26,scan_cap=416 + ksum:m=4)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.